

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (EC) Examination

December 2012

EC - 703 Digital System Design

Date: 20.12.2012, Thursday Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 70

Instructions:

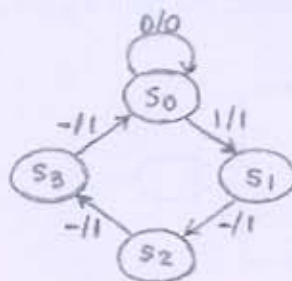
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

- Q-1 (a) What is the main difference between a PAL- and a GAL-based SPLD? 2
- (b) What are aliases? Explain with example. 3
- (c) An N-bit binary up-down counter is to be realized using a 22V10. The counter has control inputs U and D and a clock input. (U=1, count up; D=1, count down; U=D=0, no count; U=D=1 is not allowed.) What is the maximum value of N? Show how you arrived at your answer. 3
- (d) Explain the importance of setup and hold time in logic designing. 3
- Q-2 (a) Design and explain 4 x 4 Array Multiplier. 8
- (b) In which ways does the VHDL facilitate verification of digital design? 4

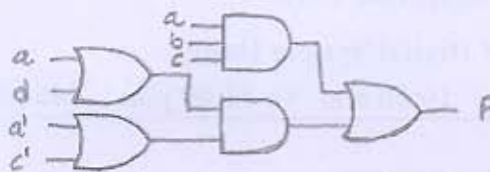
OR

- Q-2 (a) Control state graph for Serial Adder is given below. Design a control circuit for the same. 6



- (b) Design an odd parity checker. Provide the complete design flow and VHDL code with all required figures. 6
- Q-3 (a) Design a n/w which divide 6 bit dividend by 3 bit divisor. Convert it to SM chart and implement it with PLA and flip-flops. 9

- (b) Find static 1 hazard in the following network. For each hazard, specify the values of the input variables and which variable is changing when hazard occurs. 3

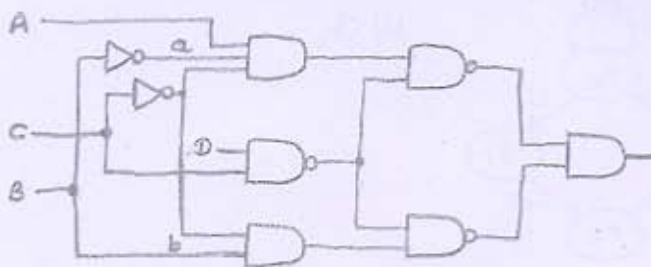


OR

- Q-3(a) Draw and explain the SM chart for UART transmitter. 6
- (b) Using the method of map entered variables, use 4 variable maps to find a minimum sum of product expression. Also verify it with 5 variable kmap. 6
- $F(A,B,C,D,E) = \sum m(0,4,5,7,9) + \sum d(6,11) + E(m_1, m_{15})$
- $F(A,B,C,D,E) = \sum m(0,4,6,13,14) + \sum d(2,9) + E(m_1, m_{12})$

SECTION-II

- Q-4(a) Write a model for a 2x1 multiplexer using tristate buffer only in VHDL 4
- (b) Implement a standard LFSR for the characteristic polynomial $f(x) = x^8 + x^7 + x^2 + 1$. 2
- (c) Describe a mealy finite state machine that will detect three consecutive coin tosses (of one coin) that results in heads. 3
- (d) What is burn in and production testing? 2
- Q-5 (a) For the given combinational circuit, 8
- Find the set of all tests that detect the fault b s-a-0.
 - Find the set of all tests that detect the fault a s-a-1.



- (b) Design a Moore sequential network for the traffic light controller. 4

OR

- Q-5 (a) Describe the complete design flow for programmable logic with all necessary figures and details 7

- (b) Justify the sentence "Structurally equivalent faults are functionally equivalent. But functionally equivalent faults are not structurally equivalent." 5
- Q-6 (a) What is BILBO? Describe BILBO architecture and its operation? 6
- (b) Draw and Explain the SM chart for Dice game. 6

OR

- Q-6 (a) Which are the main modules of keypad scanner? Explain the working of keyscan using state graph. 6
- (b) For given Boolean function below, minimize the function and generate PLA table. Implement these functions using PLA logic and draw necessary sketches. 6
- $F_1(A,B,C,D) = \sum m(2,3,5,7,8,9,10,11,13,15).$
- $F_2(A,B,C,D) = \sum m(2,3,5,6,7,10,11,14,15),$
- $F_3(A,B,C,D) = \sum m(6,7,8,9,13,14,15).$
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